

Multiple Choice Questions

1) (4 marks) **How many** lines of code are incorrect or missing in the generated code shown on the right based on the Umlle specification for the state machine on the left? Assume that the fields, enumeration, and other methods not shown on the right are correct.

- a) Two
b) Three
c) Four
d) Five

```
class A {
    sm {
        s1 {
            e1 -> s2;
            e2 [isActive()] /
                {activate();} -> s1;
        }
        s2 {
            e2 -> s2;
            e3 -> s1;
        }
    }

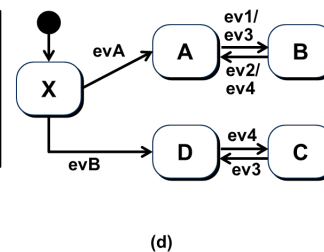
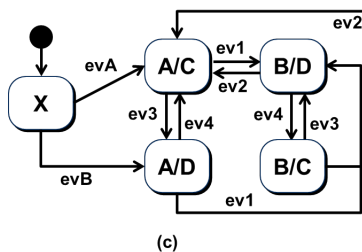
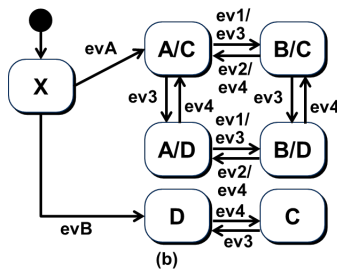
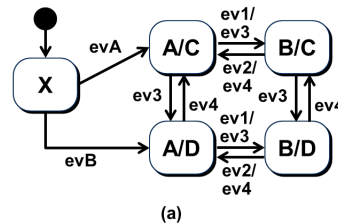
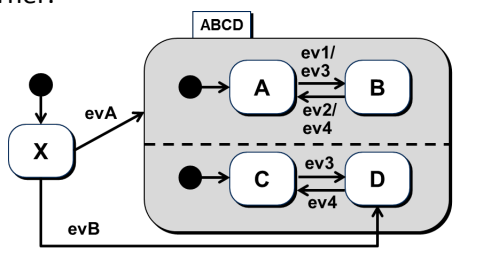
    private boolean isActive() {
        // TODO
    }

    private void activate() {
        // TODO
    }
}
```

```
01 public A()
02 {
03 }

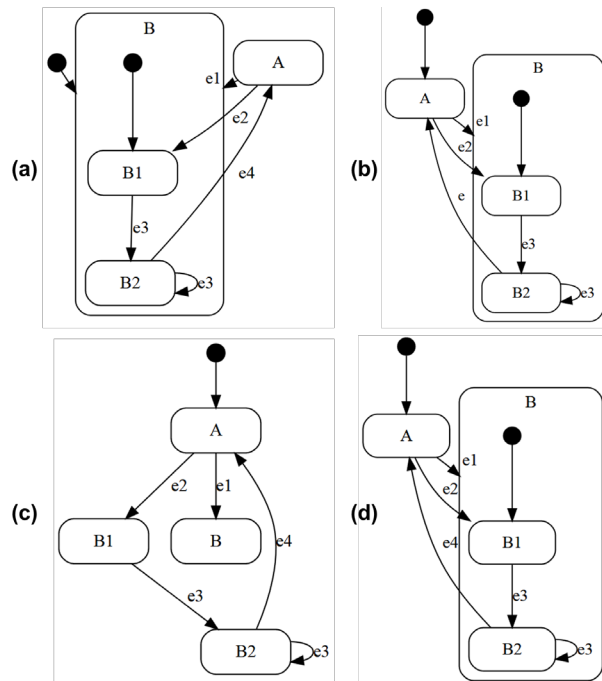
04 public boolean e2() {
05     boolean wasEventProcessed = false;
06     Sm aSm = sm;
07     switch (aSm) {
08         case s1:
09             activate();
10             if (isActive()) {
11                 setSm(Sm.s1);
12                 wasEventProcessed = true;
13                 break;
14             }
15             break;
16         case s2:
17             setSm(Sm.s1);
18             wasEventProcessed = true;
19             break;
20         default:
21             // Other states do respond to this ...
22     }
23     return wasEventProcessed;
24 }
```

2) (4 marks) Identify all of the flattened state machines that conform to the state machine in the top left corner.



3) (4 marks) Identify all of the following state machine diagrams that correspond to the Umlle specification.

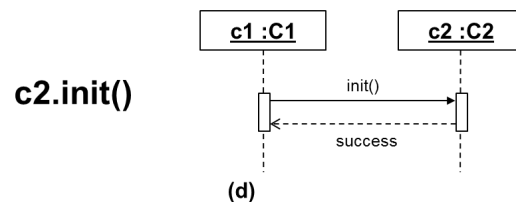
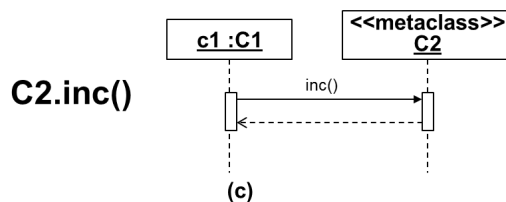
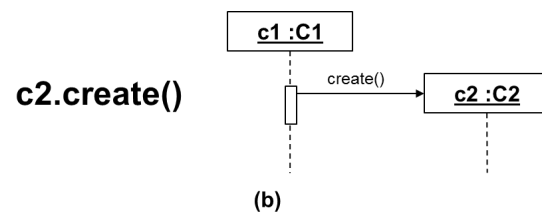
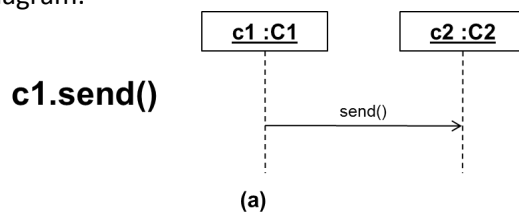
```
class C {
  sm {
    A {
      e1 -> B;
      e2 -> B1;
    }
    B {
      B1 {
        e3 -> B2;
      }
      B2 {
        e3 -> B2;
        e4 -> A;
      }
    }
  }
}
```



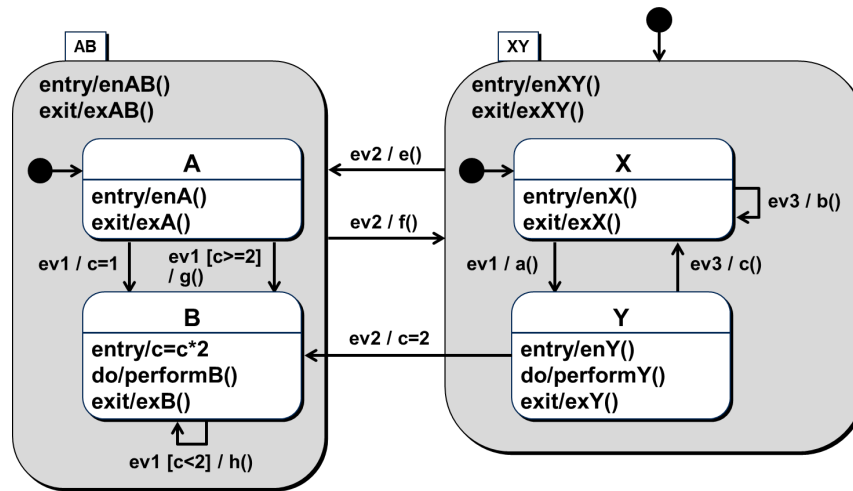
4) (3 marks) Identify all of the following statements that are incorrect.

- a) An asynchronous message in a sequence diagram has a return message while a synchronous message does not have a return message.
- b) An execution specification in a sequence diagram represents a method body.
- c) Alt, opt, loop, and par are the only combined fragments that exist for sequence diagrams.
- d) A message or combined fragment in a sequence diagram represents each and every line of code in the described method.

5) (4 marks) Identify all of the following pairs where the implementation matches the sequence diagram.



6) (4 marks) Identify all of the following action sequences that do not conform to the state machine.

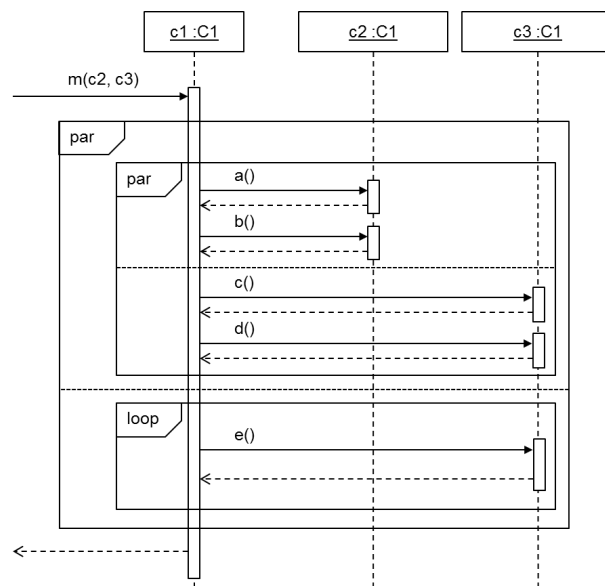


- a) enXY() — enX — exX() — a() — enY() — performY() — exY() — c=2
- b) e() — enAB() — enA() — exA() — c=1 — c=c*2 — performB() — exB() — h()
- c) c=2 — c=c*2 — performB() — exB() — g() — enA() — exA() — c=1
- d) exA() — c=1 — c=c*2 — performB() — exB() — g() — enA() — exA() — c=1

7) (3 marks) Identify all of the following statements that are correct.

- a) It is possible to define a history state in any state of a state machine.
- b) It is possible to indefinitely defer an event in a state machine.
- c) It is good practice for two orthogonal regions in a state machine to interact with each other by one region setting the state in another region.
- d) It is good practice for two orthogonal regions in a state machine to interact with each other by one region querying the state in another region.

8) (4 marks) Identify all of the following message sequences that conform to the sequence diagram. Assume that all preceding and following messages are correct.



- a) <preceded by other messages> -
e() - a() - e()- e()- b() -
<followed by other messages>
- b) <preceded by other messages> -
a() - b() - c()- d()- e() -
<followed by other messages>
- c) <preceded by other messages> -
e() - d() - e()- e()- d() -
<followed by other messages>
- d) <preceded by other messages> -
a() - e() - e()- e()- e() -
<followed by other messages>

Fill in your first and last name, McGill ID, and then date and sign.

Student Name		McGill ID											
Signature			Date										

You may detach this in-class quiz page, but you have to return it together with the other in-class quiz pages to the invigilator at the end of the in-class quiz.

Question B (70 marks)

Given the problem description below, specify the **state machine** for the dishwasher. Show the states as well as transitions including events and guards. Events and guards may require parameters. Define at least two regions for the state machine. You do not need to specify the actions of the dishwasher on transitions or in states unless needed for guards and for the communication among regions. If you use private methods for guards or actions in your state machine, use meaningful names for those methods. You do not need to define the method bodies of those methods.

Problem Description: Dishwasher

A dishwasher comes with various programs that govern how the dishwasher cleans dishes. The user may select one of the programs, adjust the drying time, and press the start button to start the selected program.

When a dishwasher is started, water is first taken from the water intake pipe. Then, the dishes are washed for 10 minutes and the water is drained. These steps are repeated a certain number of times depending on the chosen dishwasher program. Afterwards, the dishwasher dries the dishes with hot air. By default, the standard drying time of 20 minutes is used. However, the drying time may be extended to 40 minutes at any point during the dishwasher program. The drying time may also be reduced back to the default time but only as long as the drying phase has not started. When the dishwasher is active, its door is locked until the start of the drying phase. It is possible to open the door during the drying phase. When this happens, the drying phase is suspended. The drying phase continues if the door is closed within 5 minutes of opening it. If the door stays open longer, then the dishwasher does not continue with drying the dishes and ends the dishwasher program.